KEDGE dart v31: engineering report

What this covers: how the dart behaves with the launcher, in flight, on landing, in the soil and with the reel; which design versions were tried and why v31 was chosen; what the firmware and the rover client were tested against. All numbers come from the exact print geometry (scripts in source/). Assumptions are listed where they are used; the ones marked MEASURE should be replaced with bench numbers.

1. Summary

Item	Result	What it means
Length, mass	152 mm body (170 folded incl. augers), 260 g (109 printed + 152 bought)	Matches the 245 g beer-bottle target
Collision sweep	clean	kick 0 to 201.75° (203 steps), drills 0 to 43.0° (44 steps), kick x drill cross sweep, every sub-part, exact meshes
Hard stops	pods: horn about 1° past the deploy angle, fold lug about 2° past folded; kick: rest 1.5° past the plant	Tow load pushes the legs further out: it goes into the hull, not the worms or the couplers
Launcher fit	checked against base/cad/parts/launch.py over every launcher and mission state (launcher_fit.json)	Cradle, carriage, funnel, muzzle, pusher: no contact beyond the v30 dart
Flight, launch, landing, holding at 7 m/s	kedge/DART_PHYSICS31.md (dart-physics, on this geometry)	The physics rows below come from physics.py and are kept for reference
Launch	7.04 to 7.04 m/s for 15 to 20 m; 18.9 to 18.9 g over a 0.2409 m stroke; 12.06 J	Launcher spring sizing input
Landing	129.0 g if it stops in 0 cm of soil	All parts checked at this load (section 4)
Holding force	dry sand 8.3 N, moist loam 51.1 N, firm clay 332.8 N	The reel must not pull harder than this: limit the wiper current
Augers in the ground	drills 45.7 mm, foot 35.1 mm	Holding rises with depth squared in sand
Battery	1.85 h with WiFi + video on; an anchor sequence uses about 16.0 mAh	Plenty

2. Design versions tried

Each version went through the full pipeline (generate, solve the leg angles from the ground, exact-mesh sweep, physics). Belly rule: nothing outside the Ø42 cylinder below the centreline.

Version	Change	Clean	Folded len	Drill depth	CG x	Margin cal	Hold loam N	Hold sand N
v29a_base	{}	yes	154.7	29.5	84.5	-0.11	31.6	5.0
v29b_long_auger	{AUGER_L: 66.0}	yes	162.7	33.9	84.8	-0.1	32.0	6.0
v29c_cells_fwd	{LAYOUT: cells_fwd}	yes	154.7	29.5	81.5	-0.04	31.6	5.0
v29d_low_pivot	{DRILL_PIV_Z: 5.0, AUGER_L: 62.0}	NO: drill_leg_L, drill_leg_L_auger	158.7	27.9	84.7	-0.11	30.2	5.5
v29e_fwd_long	{LAYOUT: cells_fwd, AUGER_L: 66.0}	yes	162.7	33.9	81.8	-0.03	32.0	6.0
v29f_fwd_long_fat	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 8.0}	yes	162.7	36.3	82.3	-0.03	41.3	6.6
v29g_fwd_xlong	{LAYOUT: cells_fwd, AUGER_L: 72.0}	yes	168.7	37.4	82.0	-0.02	32.4	6.9
v29h_fwd_long_foot50	{LAYOUT: cells_fwd, AUGER_L: 66.0, FOOT_AUGER_L: 50.0}	yes	162.7	33.9	81.9	-0.02	35.2	6.3
v29i_piv10	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 10.0}	yes	162.7	40.7	82.3	-0.03	40.1	6.4
v29j_piv12	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 12.0}	yes	162.7	42.4	82.2	-0.03	37.5	6.0
v29k_piv14	failed to build							
v29l_piv12_d18	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 18.0, DRILL_PIV_Z: 12.0}	NO: drill_leg_L_auger | hull_top, drill_leg_L_auger | tail_cap	162.7	42.4	83.0	-0.04	35.1	6.7

Version	Change	Clean	Folded len	Drill depth	CG x	Margin cal	Hold loam N	Hold sand N
v29m_piv11	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 11.0}	yes	162.7	42.8	82.3	-0.03	39.3	6.2
v29n_piv12	{LAYOUT: cells_fwd, AUGER_L: 66.0, AUGER_D: 16.0, DRILL_PIV_Z: 12.0}	yes	162.7	45.0	82.2	-0.03	38.5	6.1
v29o_piv10_l70	{LAYOUT: cells_fwd, AUGER_L: 70.0, AUGER_D: 16.0, DRILL_PIV_Z: 10.0}	yes	166.7	43.4	82.5	-0.03	44.9	7.0
v29p_foot30	{FOOT_AUGER_L: 30.0, FOOT_TIP: 8.0}	yes	162.7	42.8	82.0	-0.05	37.5	6.0
v29q_piv8_l70	{AUGER_L: 70.0, DRILL_PIV_Z: 8.0}	yes	166.7	39.0	82.5	-0.03	41.7	7.3
v29r_piv9_l70	{AUGER_L: 70.0, DRILL_PIV_Z: 9.0}	yes	166.7	41.1	82.5	-0.02	43.2	7.1
v29s_piv10_l70	{AUGER_L: 70.0, DRILL_PIV_Z: 10.0}	yes	166.7	43.4	82.5	-0.03	44.9	7.0
v29t_piv10_l74	{AUGER_L: 74.0, DRILL_PIV_Z: 10.0}	yes	170.7	46.1	82.7	-0.02	45.3	7.6
v29u_light	{LIGHT: 1}	yes	166.7	43.4	82.4	-0.02	44.9	7.0
v29v_d17	{AUGER_D: 17.0, DRILL_PIV_Y: 31.0}	yes	166.7	43.4	82.8	-0.03	45.2	7.3
v29w_d17_l72	{AUGER_D: 17.0, DRILL_PIV_Y: 31.0, AUGER_L: 72.0}	yes	168.7	44.8	83.0	-0.02	48.5	7.7
v29x_pitch9	{AUGER_PITCH: 9.0}	yes	166.7	43.4	82.9	-0.03	45.0	7.0
v29y_l72	{AUGER_L: 72.0}	yes	168.7	44.8	82.6	-0.02	45.1	7.3

Holding columns use the final (corrected) soil model for every version; versions with pods at y 29 were swept before the 1.5 mm pod move, the final v29 sits at y 30.5.

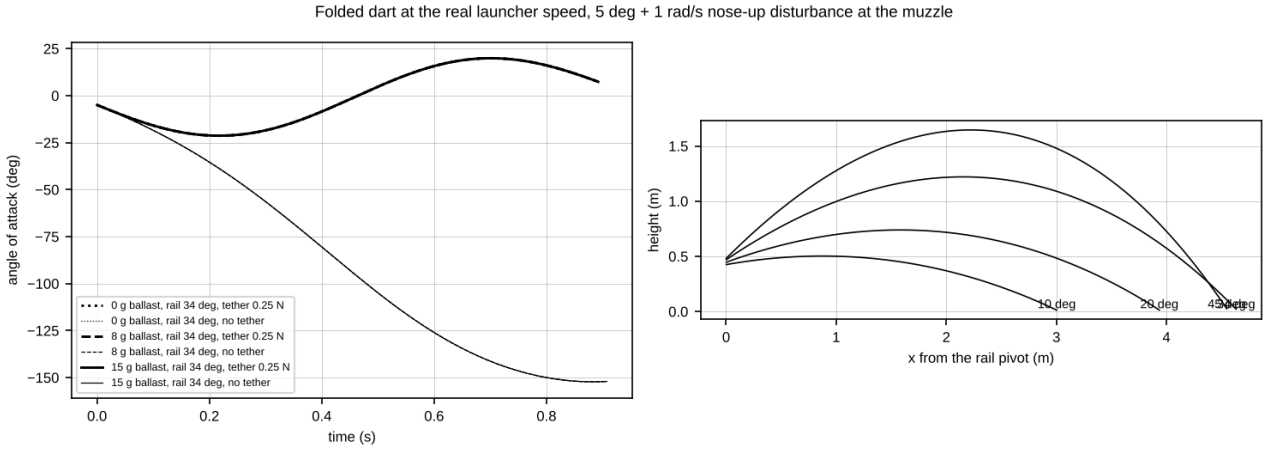
v31 (this build): v30 (Ø17 x 72) taken as the start because the launcher is built around it, then the anchoring hardware reworked: deploy stop horns and fold lugs on the hull, a kick rest on the ear root, a longer pod stack (terminal room, bushing gap, bearings back to back), drill pivot z 11.0 so the longer pods still reach 43.0°, augers Ø13 x 72.8 with a Ø6 core (dart-physics: same loam holding as Ø17, about half the install torque and a narrower cut while milling), pods at y 29. History below. Before that: v29 = cells in front of the boards, Ø16 x 70 augers, drill pivot raised to z 10, pods at y 30.5. Final: drills deploy to 43.0° and reach 45.7 mm deep (v28: 29.5); holding in loam 51.1 N against 31.6 N for the v28-style baseline. What the study showed: raising the drill pivot lets the pods swing further before their noses touch the ground, so the augers go deeper; lowering it does the opposite and breaks the belly rule. But depth alone is not holding: with the pull horizontal, a steeper auger takes more of it sideways. Embedded length matters most, so 70 mm augers at pivot 10 beat 66 mm at pivot 11 or 12. 74 mm adds 1% for 4 mm more length past the tail; Ø17 x 72 (pods out to y 31) adds 8% for 2 mm more length: the best gain left, kept as an option; a 9 mm pitch changes nothing in this model; a lightened tray and kick keel saves under 1 g and weakens the pole; pivot 14 does not fit the deploy worm; Ø18 hits the hull folded. The 30 mm foot option (10 mm lower for the barrel) costs about 2 N; the 50 mm foot (+9 mm depth) adds about 3 N.

3. Flight

Centre of mass at x = 80.9 mm from the nose (of 152); the projected-area centre of pressure is at x = 49.9 (side) / 50.9 (top). On the body alone the dart is neutral, so it was simulated: a pitch-plane rigid-body flight with the real mass, inertia and side profile; the cases and the 7 m/s launcher numbers are dart-physics' (kedge/flight_sim.py, DART_PHYSICS31.md).

Case	Max angle of attack after 0.6 s	Tumbled	Range m
pitch, 0 g ballast, rail 10 deg, tether 0.25 N	None	no	3.0
pitch, 0 g ballast, rail 10 deg, no tether	None	YES	3.15
pitch, 0 g ballast, rail 20 deg, tether 0.25 N	19.5	no	3.94
pitch, 0 g ballast, rail 20 deg, no tether	131.0	YES	4.2
pitch, 0 g ballast, rail 34 deg, tether 0.25 N	19.9	no	4.63
pitch, 0 g ballast, rail 34 deg, no tether	152.3	YES	5.08
pitch, 0 g ballast, rail 45 deg, tether 0.25 N	16.5	no	4.55
pitch, 0 g ballast, rail 45 deg, no tether	180.0	YES	5.19
pitch, 8 g ballast, rail 34 deg, tether 0.25 N	19.9	no	4.63
pitch, 8 g ballast, rail 34 deg, no tether	152.3	YES	5.08
pitch, 15 g ballast, rail 34 deg, tether 0.25 N	19.9	no	4.63
pitch, 15 g ballast, rail 34 deg, no tether	152.3	YES	5.08

Case	Max angle of attack after 0.6 s	Tumbled	Range m
yaw, 0 g ballast, rail 34 deg, tether 0.25 N	9.3	no	4.62
yaw, 0 g ballast, rail 34 deg, no tether	131.9	YES	4.62
yaw, 8 g ballast, rail 34 deg, tether 0.25 N	9.3	no	4.62
yaw, 8 g ballast, rail 34 deg, no tether	131.9	YES	4.62
yaw, 15 g ballast, rail 34 deg, tether 0.25 N	9.3	no	4.62
yaw, 15 g ballast, rail 34 deg, no tether	131.9	YES	4.62



Result (dart-physics, 7 m/s): with the tether on (0.25 N brake) the dart flies nose first at every rail pitch; without the tether it tumbles with or without ballast. v31 carries no nose ballast.

4. Launch and landing loads

Load case	Value	Unit	Check
cells pushed aft at launch (each)	4.1	N	onto the rigid plates on the rear cell-bay disc: contact never breaks
tail cap push at launch (dart + line)	59.3	N	4 radial M3 into the spigot ring, hull wall in compression: fine
cells forward at landing, soft soil (129 g at the CG, rigid-body upper bound)	44.0	N	onto the 3 mm front cell-bay disc and its springs
cells forward at landing, gravel (492 g at the CG, rigid-body upper bound)	166.0	N	onto the 3 mm front cell-bay disc and its springs
cells forward at landing, hard ground (998 g at the CG, rigid-body upper bound)	336.0	N	onto the 3 mm front cell-bay disc and its springs
coupler bending at landing, soft soil (pod 22 g at 338 g)	19.4	MPa	PPA-CF 80 MPa: margin 4.1
coupler bending at landing, gravel (pod 22 g at 1285 g)	73.9	MPa	PPA-CF 80 MPa: margin 1.1
coupler bending at landing, hard ground (pod 22 g at 2608 g)	150.0	MPa	PPA-CF 80 MPa: margin 0.5
kick leg at landing, soft soil	18.6	N	d3 steel shaft shear + ear bearing

Battery contacts follow from this: launch throws the cells aft, so the aft contacts are rigid plates and the springs sit on the forward disc. At landing the cells move forward into the springs for a few milliseconds; the firmware treats a reset as harmless (motors start off, WiFi is back in about a second). A 470 to 1000 µF capacitor across the 5 V output of the MP1584 rides through contact bounce and is recommended.

5. Anchoring, tipping and the reel

Soil	Each drill auger N	Foot auger N	Friction N	Total N
dry sand	2.6	1.0	1.4	8.3
moist loam	21.4	7.9	1.4	51.1
firm clay	151.2	55.1	1.1	332.8

Method: each auger is a short rigid pile (Broms): sand 0.5 γ D L² Kp; clay free-head solution. The drill augers are raked at 43.0°, so their embedded length is 66.9 mm for 45.7 mm of depth. The horizontal pull splits into a push along each auger (end bearing on the flights) and a sideways load; the weaker of the two limits governs. These are small anchors: length in the soil is everything. In loose dry sand the dart holds only a few newtons whatever the design; in loam or clay it holds tens of newtons. The foot's job is holding the nose down: it can resist about 5.5 N of uplift in loam.

Tipping: the tether pulls horizontally at the axis, 21 mm above the ground. At 51.1 N the nose wants to lift with 1.073 N·m; the kick foot 154 mm ahead of the drills needs to hold 6.99 N down. That is why the foot is planted first and the kick worm stays locked.

Reel interface: the reel's wiper motor can pull far more than any of these numbers. Limit tether tension on the reel side: read the BTS7960 current (IS pins) or add a load cell (you have the 100 kg cell + HX711) and stop or slip when tension passes the soil's holding value (firmware/reel/reel_tension_limit does this: speed tapers over the last 30% and stops at the limit; set it with l20). Start with 20 N, raise it while watching the dart's pitch on /status: a nose-up change of more than about 5° means it is starting to drag.

6. Motors, battery, strength

Kick: the leg family weighs 39.5 g with its centre None mm from the pivot, so gravity asks at most 0.312 kg·cm of the worm. Small worm N20s give 1 to 4 kg·cm: ample. Deploy: both pods together need None kg·cm against gravity; the real load is soil, which is why the drills spin while the pods swing (firmware auto sequence).

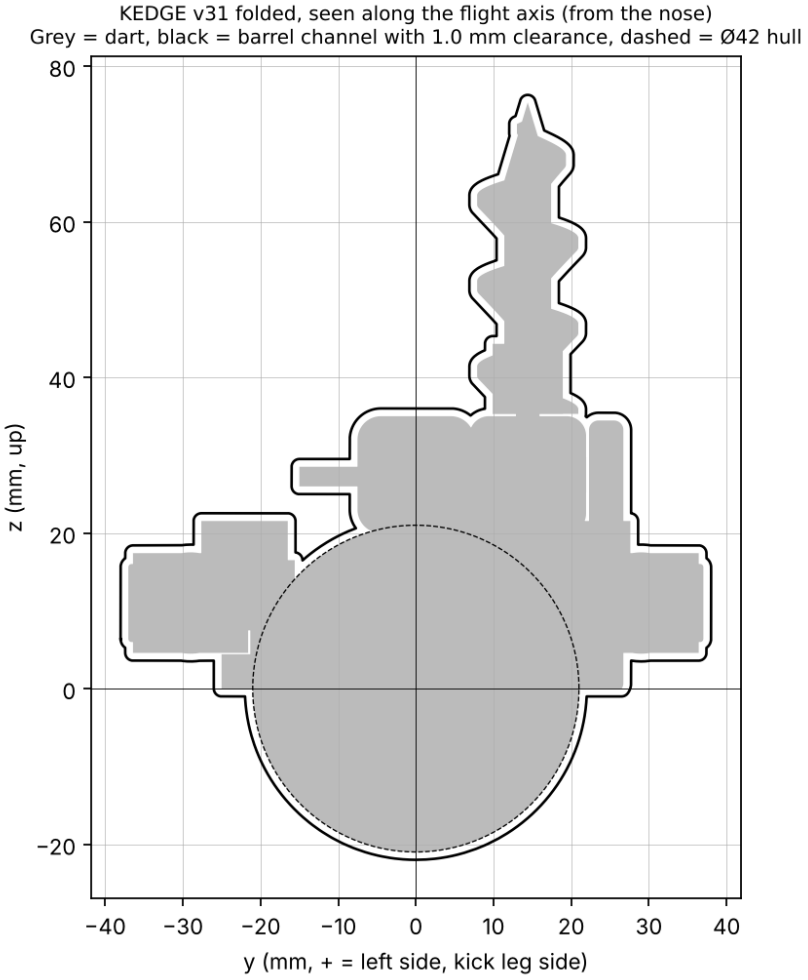
Important geometry fact: the drill augers sit radially on the pods, so their depth comes only from the swing. They cut sideways through the soil as the pods rotate; spinning them while deploying is what lets the deploy worm push them in (a slow mill: dart-physics gives the times). Under tow the soil turns the pods further down; once a pod is fully down its head sits on the hull horn, so that moment never reaches the deploy worm, its D stubs or the couplers. The same holds for the kick on its rest. Below full deploy the worm holds and the couplers are the weak link (they are cheap to reprint and protect the gearbox). The foot auger is different: near 180° its axis is almost vertical and the swing drives it along its own axis, so it really screws in.

Battery: 5.18 Wh. WiFi + VTX + camera draw about 2.52 W: 1.85 h. Worst case all motors stalled is about 2 A (BMS 8 A, INA219 trip set at 2.5 A).

Coupler torque capacity about 1.01 N·m (PLA, derated for the flat and layer lines): well above the deploy worm. Kick pole (round pole + keel, section modulus 99.8 mm³ from the real cross-section): 14.4 MPa at a 20 N sideways load on the foot; keep sideways loads under about 31.2 N (factor 2 on PLA). The pole prints lying down, so the bending runs along the extrusion lines.

7. Launcher interface

The barrel must clear this silhouette (folded dart seen along its axis). Channel with 1.0 mm clearance spans y -38.0 to 38.0, z -22.0 to 76.3 mm. Files: barrel_profile.dxf (sketch it in SolidWorks or Fusion and extrude), barrel_keepout.stl (subtract it from the barrel body), barrel_profile.svg, and KEDGE_v31_assembly_folded_STEP.zip (the whole folded dart as one STEP assembly). The real launcher (base/cad/parts/launch.py) was checked directly with this dart: launcher_fit.py intersects the folded dart with every launcher part over the cock sweep, the shot, the reset, the dart leaving at 6 and 47° and every mission state with the dart on, off or coming back to the launcher.



the belly (z < 0) is the plain Ø42 cylinder over x 20..141: cradle it there; the barrel's lower half can be a plain Ø44 half-pipe. push on the tail cap end face (flat ring Ø42, tether holes in the middle): keep the pusher face a ring so the tether exits through its centre.

The folded foot auger is the tallest thing on the dart (top of the silhouette). If the barrel slot for it is a problem, the options in order of cost are: a shorter foot auger (30 mm: about 10 mm less height, about 8 mm less depth), or a foot auger tilted 45° toward the tail in the housing (folded it leans back over the hull; deployed it points down and forward, which resists the tether better). Neither is built; both are quick to generate on request.

Spring energy needed: about 12.06 J delivered to the dart for 20 m, more with losses. A longer stroke lowers the peak load on the dart: 0.4 m halves the g.

8. Firmware and rover link

The control logic is a plain C++ header (kedge_logic.h) shared by the sketch and a PC test that drives it against a motor model. The auto sequence is: kick swings out; when the kick current rises (foot auger touches) the foot auger starts; the kick keeps pushing until the housing lands, read by pausing the foot for 150 ms every 400 ms so the INA219 sees the kick alone; the foot finishes; then the pods deploy with the drills spinning, deploy stall read the same way with the drills paused; drills finish; hold.

Rover API on the KEDGE WiFi: GET /status (JSON: rail volts, current, peak, motors, sequence, armed, landed, roll, pitch, last event), GET /key?c=&ms;=, GET /stop, GET /auto. kedge_client.py (standard library only) gives status, watch-to-CSV, hold-a-key, auto with progress, terminal driving, and an FPV window with the telemetry over the RC832 + capture card picture (OpenCV).

Landing detection: the BNO085 accelerometer sees the launch spike, flight, impact and stillness; with A (arm) pressed before launch the dart anchors itself 1.5 s after it comes to rest.

Test results (PC simulation of the real logic, and the client against a mock dart):

ok: 8 motor pins + SDA/SCL are 10 different GPIOs

ok: only DRV#1 and DRV#2 are used: DRV#3 is off the dart

ok: DRV#1 A = deploy, B = kick (unchanged)
ok: DRV#2 channel A = both drill motors
ok: DRV#2 channel B = foot
ok: every motor has its own driver channel
ok: Dn labels match the XIAO ESP32S3 GPIO numbers (SDA D4, SCL D5)
ok: reset-twitch pins D6/D7 only on the slow foot worm, drills on D8/D9
ok: drill pair stall 700 mA fits one 1.2 A channel; drill stall threshold below it
ok: trip above all four stalled (1610 mA) and inside the INA219 3.2 A range
ok: flight log RAM: 600 x 20 B samples, under 12.3 kB
ok: boot: all motors off, phase IDLE
ok: boot after brownout: motors off, phase FAULT (BROWNOUT)
ok: stop clears the boot fault
ok: boot after watchdog reset: phase FAULT (WATCHDOG)
ok: auto accepted from IDLE
0 ms boot, motors off
0 ms auto: kick swing
3170 ms auto: foot touches, foot auger on
5330 ms auto: kick down, foot finishing
6830 ms auto: deploy with drills spinning
9290 ms auto: pods down, drills finishing
12290 ms auto: anchored, holding (worms self-lock)
13290 ms battery ok (7.60 V at rest)
ok: auto sequence reaches anchored, phase ANCHORED
ok: foot auger starts when the foot meets the ground (kick current rise)
ok: kick stops after the housing lands (stall read with the foot paused)
ok: deploy starts only after the foot is planted
ok: deploy stops on its own stall although the drills draw more
ok: drills finish 3 s after the pods stop
ok: everything off at the end
ok: phases IDLE > ANCHORING > ANCHORED
ok: auto refused when already anchored
ok: manual keys refused while auto runs
ok: release refused while auto runs (busy)
ok: repeated /auto while anchoring: accepted, not restarted
ok: space aborts the auto sequence, everything off
ok: drills stalling in the finishing phase stop early and count as anchored
ok: kick never meets the ground: KICK_TIMEOUT fault, all off
ok: release accepted from ANCHORED
ok: release step 1: drills unscrew alone
ok: repeated /release accepted without restart; /auto refused while releasing
ok: release step 2: pods lift while the drills keep unscrewing (no deploy-alone lead: augers may still be in soil)
20000 ms release: drills out
23000 ms release: pods up, drills still out
24770 ms release: pods up, foot out
26270 ms release: kick back, foot still out
31540 ms release: done, legs stowed
ok: pods stop on the stowed-end stall
ok: foot unscrews alone 1.5 s, then the kick swings back
ok: kick stops on the stowed stall (read with the foot paused)
ok: released: legs stowed, all motors off
ok: phases RELEASING > IDLE
ok: status seq 'released'
ok: drills jammed on the way out: rocked 5 times, then DRILL_JAM, all off
ok: foot jammed on the way out: FOOT_JAM, all off
ok: kick never reaches the stow stop: KICK_TIMEOUT within 15 s
ok: whole-sequence time limit: SEQ_TIMEOUT, all off
ok: release from IDLE with legs already stowed still ends (stall at once)
ok: WiFi hold: repeats keep the kick running
ok: WiFi hold: key released, stops within the 350 ms deadman
ok: serial key runs 1.5 s
ok: serial key stops after 1.5 s
ok: manual run capped...
ok: ...at 8 s whatever ms was asked
ok: held key (repeats every 150 ms) stops at the 8 s hold cap
ok: ...and stays off while the key is still held
ok: released and pressed again: runs
ok: no current sensor: a held key stops after 2 s
ok: stall threshold scales with speed (kick stall still caught at speed 155 in the plant)
ok: sequences always run at full duty
ok: two manual motors: one stalled (deploy in its end + drills working) is caught
ok: held key into an end stop: stall stops it although repeats keep coming
ok: after the key is released it may be tried again
ok: after a stop the other direction is never locked
ok: deploy driven into its end: stall stop
ok: WiFi client lost: manual motors stop at once

ok: WiFi lost during auto: the sequence carries on
ok: unknown key refused
ok: over-current trip: all off, FAULT OVERCURRENT
ok: stop clears the fault
ok: one 5 ms current spike does not trip
ok: rail under 5.6 V for 0.5 s while driving: LOW_RAIL, all off
ok: low rail while idle is not a fault
ok: INA219 lost during auto: CURRENT_SENSOR fault, all off
ok: auto and release refused without the current sensor
ok: run keys refused while the fault is latched
ok: after stop (and the key released), manual keys work without the current sensor
ok: ...and still stop on the deadman
ok: INA219 readings back: sensor OK again
ok: one or two failed reads are tolerated
ok: arm: phase ARMED
ok: 4+ g for 10 ms: still ARMED
ok: launch: |a| over 4 g for 20 ms, launch time = start of the push
ok: auto refused during FLIGHT
ok: motor keys refused during FLIGHT
ok: release refused during FLIGHT
ok: still FLIGHT just after the impact
ok: landing: impact over 6 g then still 1 s; landing time = the impact
ok: armed with auto: anchoring starts by itself on landing
ok: status JSON has phase and t_launch_ms
ok: phases ARMED > FLIGHT > ANCHORING > ANCHORED
ok: flight log frozen
log window 1005..7005 ms, launch 3000, land 4010, 301 samples
ok: log starts 2 s before launch detection
ok: log ends 3 s after landing, then frozen
ok: about 50 Hz: (2 + 1 + 3) s window = about 300 samples
ok: flightlog JSON shape, no chunk overflow
ok: flightlog says how it landed
ok: frozen log ignores new samples
ok: arming while ANCHORED is refused (release first)
ok: ...and still refused after a key tap with the legs out ('legs out: release first')
ok: re-arm after the release
ok: re-arming unfreezes the log
ok: a 15 ms knock over 4 g is not a launch
ok: not armed: a launch trace is ignored
ok: soft landing without an impact spike: landed after 2 s still
ok: log-only arm: phases ARMED > FLIGHT > LANDED, no auto
ok: tether jerk and tumbling: landing time = the last impact before stillness
ok: IMU silent in flight: landing assumed after the 15 s flight timeout
ok: long saturated push (100 ms) is not taken as the impact
ok: stop with disarm=0 keeps flight tracking and the arm
ok: stop during FLIGHT ends flight tracking and disarms
ok: ...and the log freezes 3 s later
ok: manual keys work again after stop
ok: armed? no; bench handling never triggers
ok: armed: repeated 10 ms knocks to 4.5 g never trigger
ok: arm refused without the BNO085
ok: arm with auto refused without the INA219, log-only arm allowed
ok: key A arms
ok: key A again disarms
ok: 50 Hz grid from a 200 Hz feed
ok: ring holds 600 samples (12 s)
ok: ring spans 12 s, oldest dropped
ok: samples in time order across the wrap
ok: not frozen: whole live ring, t_launch_ms null
ok: quaternion stored to 1e-4
ok: acceleration stored to 0.01 m/s^2 (8 g = 78.5 fits)
ok: out-of-range values clamp, no wrap
ok: uneven 7 ms feed still logs 50 per second
ok: after a gap the grid restarts at once
ok: 10 s flight: window longer than 12 s keeps 2 s before launch and the first 10 s (launch never lost)
ok: small 256 B chunks never overflow
ok: empty log is valid JSON with no samples
ok: NaN / inf inputs never reach the status JSON
ok: arming while ANCHORED is refused (legs out, release first)
ok: after the release, arming starts a new shot (phase ARMED)
ok: ...and auto runs on the next landing
wrap: 301 samples
ok: launch just after the millis() wrap keeps the full 2 s before launch
ok: foot stalls while the kick plants: foot stopped
ok: ...within 1.5 s

ok: drills stall while the pods swing: drills stopped, pods carry on
ok: ...and the sequence still anchors
ok: drills jam while the pods lift: DRILL_JAM, all off
ok: foot jams while the kick swings back: FOOT_JAM, all off
ok: armed dart dropped on a cushion: not a launch at all (free fall before the hit), auto NOT started
ok: armed dart knocked hard on the bench: auto NOT started
ok: 15 ms push over 4 g: no launch
ok: 20 ms push over 4 g: launch
ok: 40 ms push over 4 g: launch
ok: landed, auto stopped by hand: phase LANDED (not IDLE)
ok: base launcher shot (28 g, 39 ms push, 0.8 s flight): launch, landing, auto
ok: key repeats every 400 ms (slower than the 350 ms deadman): stall still stops it
ok: key repeats every 400 ms: the 8 s hold cap still applies
ok: battery at 6.4 V: auto anchors (thresholds scale with rail voltage)
ok: battery at 6.4 V: release ends stowed
ok: battery at 7.4 V: auto anchors (thresholds scale with rail voltage)
ok: battery at 7.4 V: release ends stowed
ok: battery at 8.4 V: auto anchors (thresholds scale with rail voltage)
ok: battery at 8.4 V: release ends stowed
ok: 1 failed current read in 3: a stalled kick still stops within 1.5 s
ok: manual kick + drills in air: the stalled kick is stopped, the drills keep running
ok: manual kick + foot: the stalled kick is stopped, the foot keeps running
ok: manual kick + foot both working (no stall): both keep running
ok: tumbling flight (1.1 to 1.7 g): impact landing, auto starts
ok: fault while a key is held: the held key does not restart the motor
ok: ...exactly one LOW_RAIL trip during the hold
ok: an IMU timestamp 1 ms ahead of the tick does not end the flight (no unsigned wrap)
ok: drill current drops 20 % while the pods pause: pods still go fully down
ok: foot current drops 40 % while the kick pauses: kick still plants fully
ok: foot current rises while the kick pauses: plant time limit, still anchors, no fault
ok: ms=500 repeats every 560 ms into an end stop: stall stop holds (window = ms + 250)
ok: ...and the 8 s hold cap holds
ok: +/-15 mA read noise at speed 125: held kick into its stop still stalls out
ok: armed dart dropped 20 cm onto a desk with a bounce: auto NOT started
ok: boot fault latched: run keys refused until stop
ok: a manual motor running at launch is stopped
ok: kick wire open (or shorted, driver chopping): NO_CURRENT fault within 1 s, all off
ok: manual key on an open drill wire: stopped with a message, no fault latched
ok: free-running kick at speed 205 is not mistaken for an open wire
ok: impact read at only 5 g (sensor filter): still an impact landing, auto starts
ok: status peak_g holds the highest |a| since arming
ok: manual at speed 255 drives full duty (slow decay with a DC input)
ok: one failed current read at a step end does not abort the release
ok: both directions of one motor held: stopped and kept off
ok: lowest speed: a free-running kick is not taken as stalled
ok: a key tapped in FLIGHT and its keyup stop (disarm=0) do not cancel auto on landing
ok: landed, then a manual motor runs: phase IDLE, not LANDED
ok: ...and LANDED again when it stops
stowed release: pods step 800 ms, kick step 1610 ms
ok: release with legs already stowed: each worm stops within 2.5 s
retry: kick off after 405 ms
ok: auto retried with the kick already planted: kick stopped within 1 s, foot planted alone, anchored
timeout landing: tLaunch 5000 tLand 6550 how timeout, log 2985..14965
ok: timeout landing after an impact: the frozen log still holds 2 s before launch through the impact
ok: landing 7000 ms after launch: frozen log keeps 2 s before launch (first 2985, launch 5000) and the impact
ok: landing 8500 ms after launch: frozen log keeps 2 s before launch (first 2985, launch 5000) and the impact
ok: landing 10000 ms after launch: frozen log keeps 2 s before launch (first 2985, launch 5000)
ok: R1-B #1: '1' and '!' alternating every 75 ms: stopped, not driven for 20 s
ok: a stop also locks a key another client keeps repeating
ok: ...until that key is released
ok: /auto while ARMED (launch never seen): disarms and starts
ok: arming with auto refused on a flat battery (6.3 V at rest)
ok: ...a log-only arm is still allowed (no motors)
ok: motor rail reading 0 V while driving: LOW_RAIL
ok: an arm expires after 15 min without a launch
ok: armed dart dropped 1.2 m with a bounce: not a launch
ok: deploy wire open: NO_CURRENT at the start of the pods step
ok: kick jams half way back: KICK_JAM (not 'released')
ok: kick stalled near stowed at the start of /auto: KICK_JAM, not 'kick already down'
ok: clean anchor: no warnings
ok: drills stopped by a stone: status warnings list says so
ok: vibrating rail (0.18 g rms) and a 0 ms ramp into the push: launch seen, auto runs
ok: vibrating rail (0.18 g rms) and a 3 ms ramp into the push: launch seen, auto runs
ok: vibrating rail (0.18 g rms) and a 8 ms ramp into the push: launch seen, auto runs
ok: vibrating rail (0.18 g rms) and a 15 ms ramp into the push: launch seen, auto runs

ok: vibrating rail (0.18 g rms) and a 25 ms ramp into the push: launch seen, auto runs
ok: a single 2.5 g sample before the push does not block the launch
ok: anchored
ok: reboot with the kick planted, then /stop and /auto: no false KICK_JAM (position unknown after a non power-on reset)
ok: near ground and a hidden housing stall (plant time limit): release ends 'released', no false KICK_JAM
ok: page key release (keep=1) then the same key 300 ms later: runs (no lock)
ok: STOP during a deadman gap still locks the repeating key
ok: STOP of a sequence locks a key that was held (refused) during it
ok: retry after an abort with the pods half down: pods go fully down (no false 'already down')
ok: stop 6.5 s into a long flight: the frozen log still starts 2 s before launch
ok: kick jams on the way back at 3800 of 4300 (stall current): KICK_JAM, not 'released'
ok: kick jams on the way back at 3400 of 4300 (stall current): KICK_JAM, not 'released'
ok: kick jams on the way back at 3000 of 4300 (stall current): KICK_JAM, not 'released'
ok: normal release after a normal anchor: no false KICK_JAM
ok: a fault while armed disarms (landing cannot clear it and drive motors)
ok: ...motors stay off
ok: drills stalled while the pods swing are not driven again in the finishing step
ok: a brief opposite-direction tap does not clear the stall lock of the held key
ok: STOP with both directions of a motor being pressed locks both
ok: an old '@' lock does not refuse a fresh '@' right after '2' presses (per-direction lock)
ok: a refused key does not change seq (still anchored)
ok: kick swings its whole travel through air to its end stop: NO_GROUND, foot, pods and drills never driven
ok: ...and /release brings the kick back (released)
plant ended with the kick at 5800 of 5800
ok: a foot current drop of 55 mA in the kick's pause does not fake a kick stall (kick reaches the housing)
ok: ...and the anchor completes
ok: reversing a held key without a stop reverses the motor (no 'both directions held' latch)
ok: both kick keys repeated together still lock the motor
ok: arm (even log-only) is refused while ANCHORED
4520 ms kick travel from stowed 4310 ms (at 8.0 V)
ok: a kick held out in air to its end stop reports its travel (within 5 % of the real 4300 ms)
ok: a kick jammed part way out in air: KICK_JAM, not 'already down' (augers never run)
ok: after NO_GROUND, a '2' tap at the end stop does not make the next /auto anchor in air
ok: travel unknown (after a reset): a retry after NO_GROUND is NO_GROUND again
ok: the 8 s hold cap refuses with 'hold limit', not 'stalled' (a client can tell it from an end stop)
ok: pods jam at 1300 of 1400 on the way up: PODS_JAM, not 'legs stowed'
ok: pods jam at 1000 of 1400 on the way up: PODS_JAM, not 'legs stowed'
ok: pods jam at 700 of 1400 on the way up: PODS_JAM, not 'legs stowed'
ok: a normal release still ends released (no false PODS_JAM)
ok: kick jams 1600 ms into its 4300 ms return: KICK_JAM (travel counted only up to a reading under the line)
ok: kick jams 1700 ms into its 4300 ms return: KICK_JAM (travel counted only up to a reading under the line)
ok: an anchor stopped with the pods down: arm refused ('legs out: release first')
ok: after a release, a key tap does not bring back phase LANDED
ok: a key held through a sequence starts a fresh hold afterwards (no instant 'hold limit')
ok: a key pressed 25 days ago does not lock that key at the next stop (old presses expire)
ok: a run key during ANCHORING is refused (busy)
ok: ...and the page's key-release stop does not abort the anchor
ok: the key-release stop stops a manual motor
ok: a retried /auto after NO_GROUND is NO_GROUND again (augers never run in air)
ok: an /auto stopped half way through the air and retried: NO_GROUND
ok: a retried /auto after a real plant still takes 'kick already down' (no false NO_GROUND)
alternating 1 / ! for 18 s: driven into the stop 4095 ms
ok: alternating 1 and ! every 300 ms: the hold cap counts per motor, so the stop is not driven without limit
ok: boot switches the video on (D10 high)
ok: 330 ms flight after a 50 ms push: impact landing (not the 'still' fallback)
ok: 370 ms flight after a 50 ms push: impact landing (not the 'still' fallback)
ok: 600 ms flight after a 50 ms push: impact landing (not the 'still' fallback)
ok: tumbling 0.35 s flight: impact landing starts the auto anchor
ok: an 'impact' 150 ms after launch (under minFlightMs 250) is not a landing impact: no auto
ok: belly down: orient upright, tilt 0 in /status
ok: on its side: /auto refused 'not upright: on its side', nothing moves
ok: on its side: /auto?force=1 starts anyway
ok: upside down: /auto refused 'not upright: upside down'
ok: nose down 60 deg (buried nose): upright, /auto runs
ok: BNO085 silent: orientation unknown, /auto is not blocked by it
ok: imuUpAxis -1: chip -x up reads upright
ok: impact landing on its side: auto not started, phase LANDED, event says why
ok: 6.7 V at rest: battery low warning, /auto still runs
ok: 6.3 V at rest: battery critical, /auto refused 'battery low'
ok: battery critical switches the video off
ok: battery critical: /release still runs
ok: hysteresis: 6.45 V after critical stays critical (0.1 V)
ok: 6.6 V after critical: low
ok: 7.4 V: ok again
ok: a sag while a motor runs is not counted as the resting battery

ok: rail 0 V (plug out, USB only): batt none
soil from 600 of 1400: pods ended at 1400, events: milling 1
ok: pods stalled in soil at 600 of 1400: milled down to the stop (not taken as down)
ok: no soil resistance: no milling, anchors as before
ok: soil_mA: the drills' current in soil is reported (about 380 mA here)
ok: drill pair near its stall: the deploy pauses 0.5 s so the cut relaxes; the step still ends (time or progress limit)
ok: release: pods held part way up are milled first, PODS_JAM after 8 s without progress
ok: roll 38 deg (leaning on a pod, the usual rest on hard ground): leaning
ok: leaning: /auto first swings the pods with the drills off
ok: after levelling the kick swing starts
ok: ...and the anchor completes
ok: still not upright after levelling: NOT_UPRIGHT fault, all off
ok: roll 75 deg: on its side, refused
ok: nose down 30 deg (window -25..+15): refused 'nose down'
ok: nose up 20 deg (in the launcher cradle, say): refused 'nose up'
ok: pitch_g in /status reads 20
ok: anchored and still: moving false
ok: anchored, tilted 4 deg by the tow: moving true, event 'anchor moving'
ok: /video?on=0 switches it off
ok: arming switches the video back on (the flight is filmed)
ok: status JSON has video, batt, batt_V, orient, tilt_deg and stays under 1000 bytes
ok: IDLE > ARMED > FLIGHT > ANCHORING > ANCHORED > RELEASING > IDLE
Compiled for XIAO_ESP32S3 on ESP32 Arduino core 2.0.17 and 3.3.12 (USB CDC On Boot: Enabled; the sketch refuses to compile with it disabled, because Serial would then take D6/D7).
Independent reviews: two separate code reviews of the firmware and wiring and one of the geometry checks and physics were run by agents that had not seen the work being produced. They found, among others: 3V3 must go to the DRV8833 SLP pad, never VCC/VIN; D6/D7 toggle at reset (drills moved to D8/D9); safety logic must not wait on the web server (now its own 5 ms task); Shift-release and touch-scroll could leave a motor running on the web page; the clay holding and the pole strength formulas were wrong; the sweep step was too coarse for the leg tip. All were fixed and re-checked.

9. Open items to measure

• Weigh every bought part on the kitchen scale and update BOUGHT_G (dart-physics model; the mass and CG use catalogue values).
• N20 end-cap tabs: the pods leave 1.2 mm (tabs bent up), the saddle and foot housing 1.6 mm: check on the first motor; the gearbox output bushing gets 0.6 mm before the shaft nut.
• Bench stall currents of each motor at 8 V: set kickContactMa / kickStallMa / deployStallMa in kedge_logic.h.
• Launcher stroke and spring: replace STROKE_M and re-run physics.py for the real g loads.
• A pull test in your soil: plant the dart, pull the tether with a luggage scale, compare with section 5.
• BMS stick dimensions when it arrives; F693ZZ flange size (catalogue 9.5 x 0.6 used).